

Aditya Kumar Pattanayak

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Education

- **International Institute of Information Technology, Bhubaneswar** *Aug 2023 - July 2027 (Expected)*
Bachelor of Technology in Information Technology **CGPA: 7.87/10.0**
- **DAV Unit 8, Bhubaneswar** *Jun 2020 - May 2022*
Senior Secondary (XII) Percentage: **94.8%**

Experience

- **Backend Engineer** *Oct 2025 - Jan 2026*
Stealth Startup (Remote)
 - Built 0→1 AI-powered market and founder-intelligence platform serving 100+ clients: real-time signal feeds, LinkedIn RAG vector indexing, and automated company research reports.
 - Architected AWS infrastructure (ECS, ECR, S3), Apify enrichment pipelines, and full-stack React/TypeScript + backend services; Dockerized services with CI/CD for production reliability.
 - Integrated third-party APIs, Stripe subscriptions, and data aggregation workflows handling production-scale traffic.

Projects

- **Aegis - AI Agent Reliability Runtime** 🐙 [Link]
Python, FastAPI, OpenTelemetry, Grafana, Docker
 - Designed and deployed a middleware reliability layer between LLM planners and external tools, enforcing deterministic capability, argument, confidence, and response validation with graceful recovery.
 - Built an evaluation harness measuring correctness, consistency (Pass@K/Pass^k), failure modes, and p50/p95 latency against an unguarded baseline; reduced silent failures from 6 → 0 in an initial 22-case benchmark.
 - Added OpenTelemetry + Grafana distributed tracing across planning, validation, execution, and recovery, enabling stage-level latency diagnosis and operational debugging.
- **ChunkdUp - Persistent Memory SDK for AI Agents** 🐙
Python, PostgreSQL, pgvector, SQLAlchemy, SentenceTransformers, spaCy, Docker
 - Built a persistent memory system for AI agents with a 4-stage policy engine (STORE/MERGE/UPDATE/IGNORE), achieving 100% decision accuracy across 54 benchmark scenarios.
 - Engineered hybrid memory retrieval using BM25, semantic search, pgvector, and Cross-Encoder reranking, achieving 0.833 MRR, 0.525 precision, and 1.0 recall on benchmark data.
 - Implemented ACID-backed PostgreSQL persistence with optimistic locking and full history/audit trails for concurrent agent access; added modular extraction (regex → spaCy NER → LLM fallback) and Dockerized deployment.
- **Exora - Real-Time AI Competitive Intelligence Engine** 🐙 [Link]
Python, TypeScript, React, FastAPI, LLM APIs, SSE
 - Built and deployed a real-time competitive intelligence platform that aggregates and analyzes market signals using multiple LLMs, with streaming responses via Server-Sent Events (SSE).
 - Designed a BYOK (Bring Your Own Key) architecture supporting multi-model workflows, enabling users to connect their own LLM providers while maintaining application-level control over usage and costs.
 - Shipped the platform to real users with an end-to-end production stack spanning frontend, backend APIs, LLM orchestration, streaming infrastructure, and deployment.

Technical Skills

Languages: Python, C++, Go, TypeScript, JavaScript, SQL

AI & ML: LLMs, RAG, Embeddings, Retrieval, Reranking, Agent Systems, Evaluation, SentenceTransformers, PyTorch, TensorFlow.js, Scikit-Learn

Backend & Systems: FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Redis

Cloud & Infrastructure: AWS (ECS, ECR, S3), Docker, GitHub Actions, CI/CD, OpenTelemetry, Grafana

Frontend: React, Next.js, TailwindCSS

Achievements

- **Winner** - ETHMumbai Triple Track Challenge for PayGate.
- **Winner** - Dev Challenge Hackathon, Vibecraft featured as Project of the Week on Peerlist (200+ users).